FLEET TEAM DIGEST

737MAX-FTD-33-23001

Issue Title : Anti-Collision Light Reliability

Airplane Model	ATA	Minor Model(s)
737MAX	3344-00	-10, -7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
04/10/2023	08/11/2026	07/30/2026

Estimated Completion Date	Next Update Date
12/31/2027	12/31/2026

Revision Description

No updates for this revision.

Applicability

737 MAX Airplanes Line Number 7850 and on.

Description

Several operators had been experiencing instances of failures and unscheduled removals of the upper and lower LED Fuselage Anti-Collision Lights (ACL) Part Number (P/N) 03-7300-0001 (upper ACL) and P/N 03-7300-0002 (lower ACL). These lights were introduced to the 737 MAX fleet starting at line number 7850.

Status

Collins and Boeing conducted investigations into these failures and identified two root causes:

(1) Moisture ingress on the upper anti-collision light. Cabin humidity and condensation is entering the light and damaging the driver board.

The corrective action is to apply BMS 3-38 compound between the pressure cover and the base plate to seal the path of ingress from the cabin.

(2) Bond wire failures, which cause the light to indicate End of Life (EOL) or Open String (pre-mature LED failure). This affects both the upper and lower ACLs.

Collins Aerospace implemented a corrective action to improve manufacturing and handling processes to mitigate the bond wire failures in service.

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Collins have further identified root causes of the bond wire failures as cyclic thermal stress due to environmental temperature variation and the power pulse duration.

Collins has proposed a redesign of the ACL assembly with the following:

- Address root cause with more LEDs at lower pulse durations
- PCBs vertically installed (prevents water conglomeration)
- LEDs embedded in optical silicone to ensure low temperature variation
- Interchangeable with actual design

Interim Action

Boeing recommends operators implement the fay seal procedure from AMM Subtask 33-44-01-420-013, step (1) (d) 1) to mitigate moisture ingress issues on the upper ACL.

If installed lights indicate early End of Life (EOL) or a pre-mature LED failure, operators should remove the ACL assembly and replace with units that have implemented the improved manufacturing and handling processes.

Final Action

For the moisture ingression issues, Boeing has approved the fay seal design change for the upper anti-collision light. The corrective action is to apply BMS 3-38 compund between the pressure cover and the base plate to seal the path of ingress from the cabin. The ACL serial number cut-in for the fay seal application is S/N 5947314 for P/N 03-7300-0001. Boeing updated AMM Subtask 33-44-01-420-013 to include the sealant application. Collins Aerospace also updated their assembly procedures in CMM 33-40-91 (Subtask 33-40-91-440-007-A01) to include the sealant application.

Collins have determined that the manufacturing process is a cause of the bond wire issues. Collins implemented a corrective action with their sub-tier supplier to improve the PCB manufacturing and handling processes. The serial number cut-ins for the improvements are S/N 5905273 for P/N 03-7300-0001 and is S/N 5870397 for P/N 03-7300-0002.

Bond wire and fay seal solution were incorporated into production with first shipment to Boeing in July 2023.

A redesign of the ACL assemblies is in work to further mitigate the bond wire failures. This change is currently in the concept phase and will require design development and certification. Current estimates for production incorporation is Q3 2028.

Note: The reliability issues pertain to any unit with S/N format 21XXXX-X-X and the S/N 7-digit format before the cut-in serial numbers. Any ACL unit with the S/N format 2XXXXX-X-X was produced prior to the S/N 7-digit format S/N XXXXXXX due to the ACL production transferring to a new site. The serialization was changed to a new system.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Fay Seal Solution

Root Cause Established: Nov 2022 (Complete)

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Solution Selected: Dec 2022 (Complete)

Engineering Test Complete: Apr 2023 (Complete)

Collins CMM Procedure Complete: Nov 2023 (Complete)

Boeing AMM Procedure Complete: Aug 2023 (Complete)

Collins Production Incorporation: July 2023 (Complete)

LED Bond Wire Solution

Root Cause Established: Late 2022 (Complete)

Manufacturing and Handling Process Improvement: July 2023 (Complete)

ACL Redesign Corrective Action Determination: March 2026 (Complete)

ACL Light Assembly Redesign Production Incorporation: Q3 2028 (Target)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	Subtask 33-44-01-420-013		
Component Maintenance Manual (CMM)	33-40-91		
Service Letter (SL)	737-SL-33-098-A		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Lower Anti-Collision Light			03-7300-0002
Upper Anti-Collision Light			03-7300-0001

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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